Arista Access Node IND-ACC-76 Indiranagar Exchange QoS Congestion Event (Microbursts, Interface Tail Drops, QoS Queue Starvation) Troubleshooting and Resolution SOP

Node: IND-ACC-76 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_1953

Description

The failure domain of QoS congestion events (microbursts, interface tail drops, QoS queue starvation) on Arista Access nodes, specifically IND-ACC-76 at the Indiranagar Exchange, is a complex phenomenon rooted in the interplay between packet queuing, scheduling disciplines, and bandwidth allocation. In this context, the Arista Access node operates as a critical juncture, aggregating and distributing network traffic between various VLANs and subnets. The Arista EOS (Extensible Operating System) employs a sophisticated QoS framework, leveraging policies and class-based queuing to prioritize and manage traffic. However, the introduction of microbursts or an overload of packets can lead to tail drop events, where the queue's capacity is exceeded and packets are discarded, resulting in congestion and potential queue starvation for other traffic classes. This SOP aims to systematically diagnose and resolve such issues on IND-ACC-76, leveraging Arista's CLI to interrogate and manipulate the network's control and data planes.

Pre-requisite Checks

- Verify the physical integrity of the Arista Access node, including temperature, power supply, and connectivity to the Indiranagar Exchange.

- Confirm the redundancy of the link and ensure the standby link is not experiencing any issues.

- Gather historical performance data, including packet loss, latency, and throughput metrics, to contextualize the current failure.

Potential Root Causes

Cause	Probability
Misconfigured QoS policy or class-map	High
Insufficient buffer size or queue depth	Medium
Congested network conditions or microbursts	Low

Diagnostic Commands & Expected Outputs

```
Command: show queueing, Expected Output: Queueing configuration and statistics for all VLANs and subnets, Description: This command will provide an overview of the current queueing configuration, including buffer sizes, queue depths, and packet drop counts. This information
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```
will help identify potential issues with QoS policy or class-map
configuration.

Command: show policy-map, Expected Output: QoS policy map
configuration and statistics, Description: This command will display
the QoS policy map configuration, including class-map definitions,
policy-map configuration, and statistics on packet matching and
policing. This information will help identify potential issues with
QoS policy or class-map configuration.

Command: show interface, Expected Output: Interface statistics,
including packet loss and latency, Description: This command will
provide interface-level statistics, including packet loss, latency,
and throughput metrics. This information will help identify potential
issues with network congestion or microbursts.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policy or class-map configuration, Command: config policy-map, Description: Review and adjust the QoS policy or class-map configuration to ensure it is correctly defined and applied to the affected VLANs or subnets.

Step 2: Step: Increase buffer size or queue depth, Command: config queueing, Description: Increase the buffer size or queue depth for the affected VLANs or subnets to mitigate the effects of packet drops and congestion.

Step 3: Step: Implement traffic shaping or policing, Command: config policy-map, Description: Implement traffic shaping or policing to regulate the traffic flow and prevent microbursts or congestion.

Post-Check Verification

```
Command: show queueing, Expected Output: Queueing configuration and
statistics for all VLANs and subnets (resolved), Description: Verify
that the queueing configuration and statistics have been updated and
that the issue has been resolved.

Command: show policy-map, Expected Output: QoS policy map
configuration and statistics (resolved), Description: Verify that the
QoS policy map configuration and statistics have been updated and that
the issue has been resolved.

Command: show interface, Expected Output: Interface statistics,
including packet loss and latency (resolved), Description: Verify that
the interface statistics, including packet loss and latency, have
improved and that the issue has been resolved.
```

Roll-back Procedures

- Step: Restore original QoS policy or class-map configuration, Command: config rollback, Description: Restore the original QoS policy or class-map configuration to ensure that any changes made during the troubleshooting and resolution process are reversed.

- Step: Revert buffer size or queue depth changes, Command: config queueing, Description: Revert any changes made to the buffer size or queue depth to ensure that the original configuration is restored.

- Step: Remove traffic shaping or policing configuration, Command: config policy-map, Description: Remove any traffic shaping or policing configuration that was implemented during the troubleshooting and resolution process.